

Answer Key

Geometric Constructions & Rigid Transformations | 53 points | about 60 minutes

Blue = original, red = image. Total 53 points. Multiple choice 1 pt each (6). Grid problems 2 pts each (16): 2 if every vertex is right, 1 if one vertex is off or the method is right. Describe 2 pts each (6). Reasoning 2 pts each (10). Constructions 3 pts each (15): 2 for clean, accurate arcs and 1 for the explanation.

Part 1: Multiple Choice (1 point each)

1. Which of these is **not** a rigid transformation?

C. Dilation by a factor of 3

A dilation changes side lengths.

2. A reflection maps $P(-3, 4)$ to $P'(-3, -4)$. What is the line of reflection?

A. the x-axis

The y-value flips sign and x stays the same, so the line is the x-axis.

3. Rotating $(2, 5)$ by 90° counterclockwise about the origin gives:

B. $(-5, 2)$

$(x, y) \rightarrow (-y, x)$ gives $(-5, 2)$.

4. A translation maps $(-2, 3)$ to $(4, -1)$. Where does it send $(0, 0)$?

C. $(6, -4)$

The shift is $(+6, -4)$, so $(0, 0)$ goes to $(6, -4)$.

5. When you construct a perpendicular bisector, you draw arcs of the same radius from both endpoints. Why does that make the crossing points land on the bisector?

A. Equal radii make each crossing point equally far from both endpoints

Every point equally far from both endpoints lies on the perpendicular bisector.

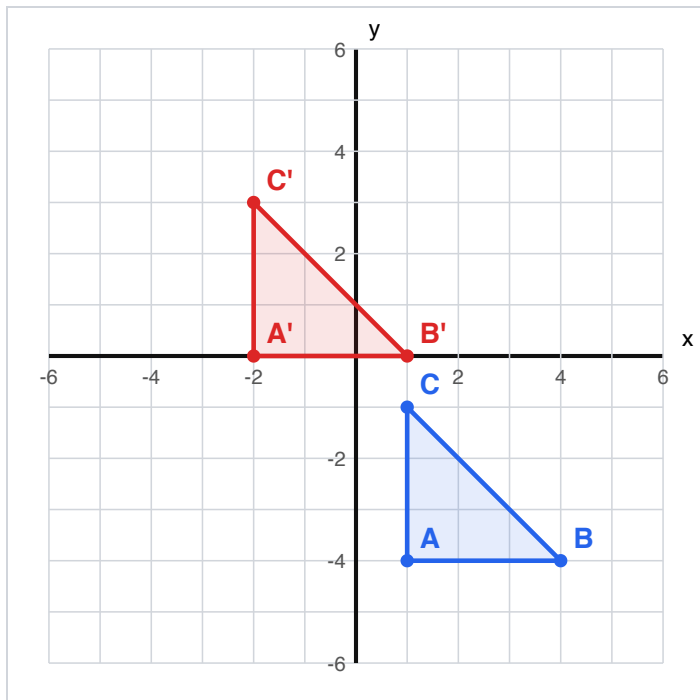
6. Triangle $A'B'C'$ is the image of triangle ABC after a reflection. Which statement must be true?

A. $AB = A'B'$

Reflections preserve distance. They reverse orientation, so D is false.

Part 2: Draw the Image (2 points each)

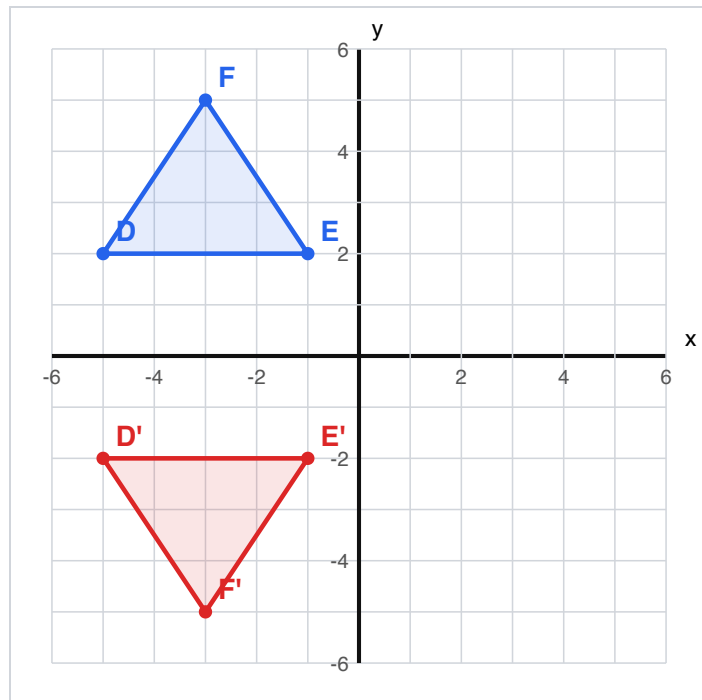
7. Translate triangle ABC by $\langle -3, 4 \rangle$.



A'(-2, 0), B'(1, 0), C'(-2, 3)

Subtract 3 from each x, add 4 to each y.

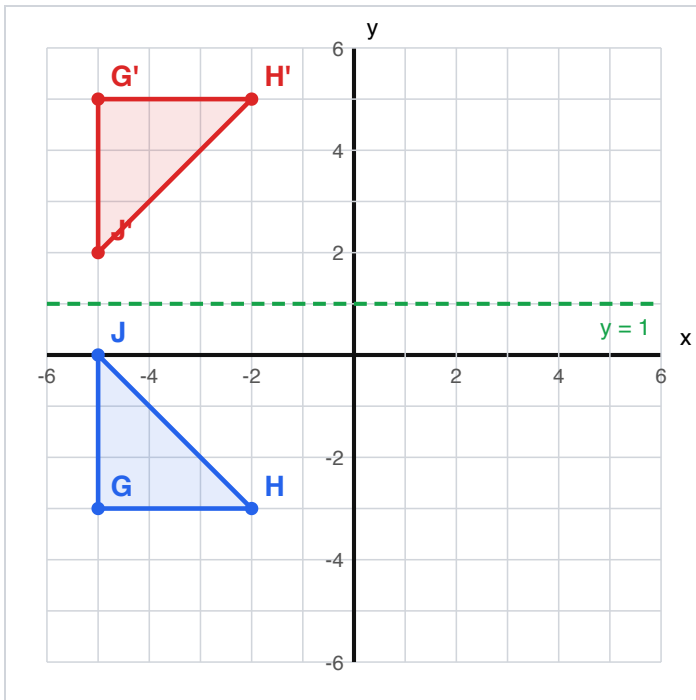
8. Reflect triangle DEF over the **x-axis**.



D'(-5, -2), E'(-1, -2), F'(-3, -5)

$(x, y) \rightarrow (x, -y)$.

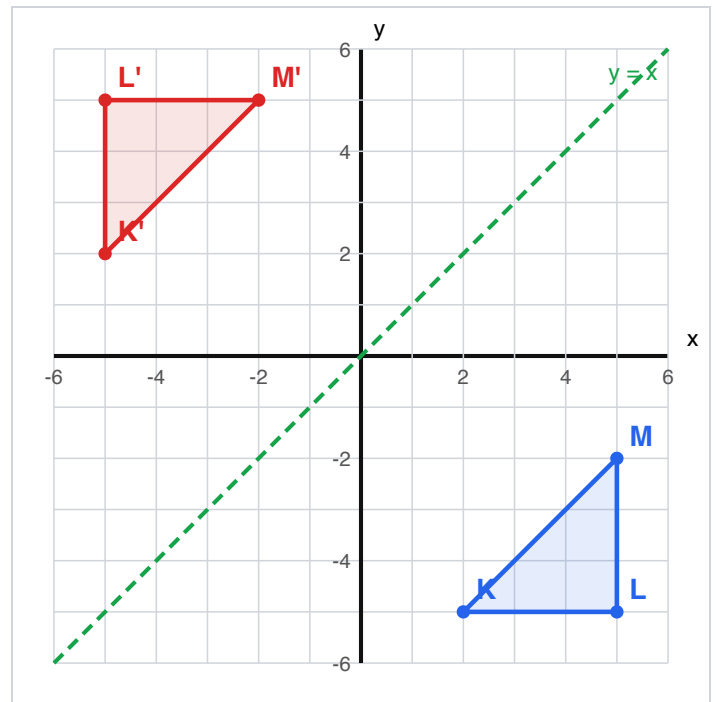
9. Reflect triangle GHJ over the line $y = 1$. (Each point and its image are the same distance from the line.)



$G'(-5, 5)$, $H'(-2, 5)$, $J'(-5, 3)$

$y \rightarrow 2 - y$. G and H are 4 below the line so G' and H' are 4 above ($y = 5$). J is 1 below so J' is 1 above ($y = 2$).

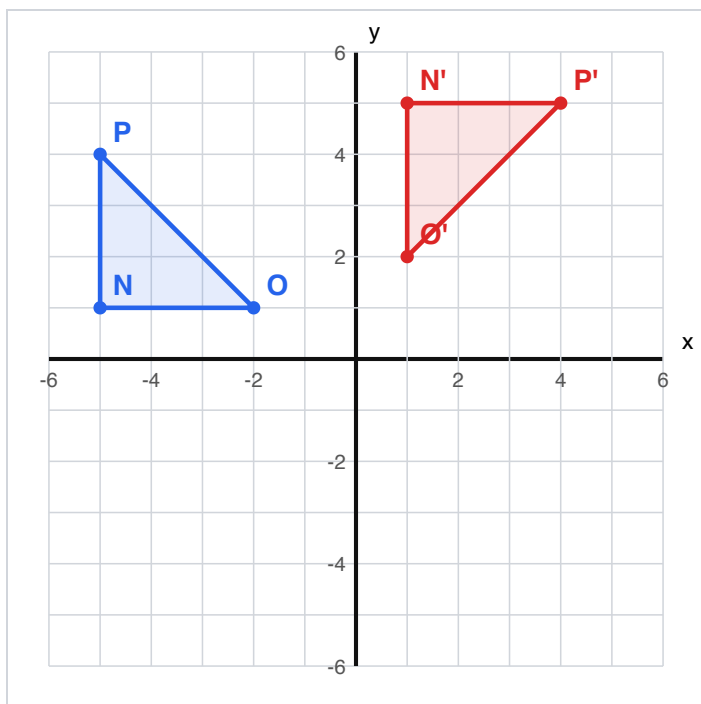
10. Reflect triangle KLM over the line $y = x$.



$K'(-5, 2)$, $L'(-5, 5)$, $M'(-2, 5)$

$(x, y) \rightarrow (y, x)$: swap the coordinates.

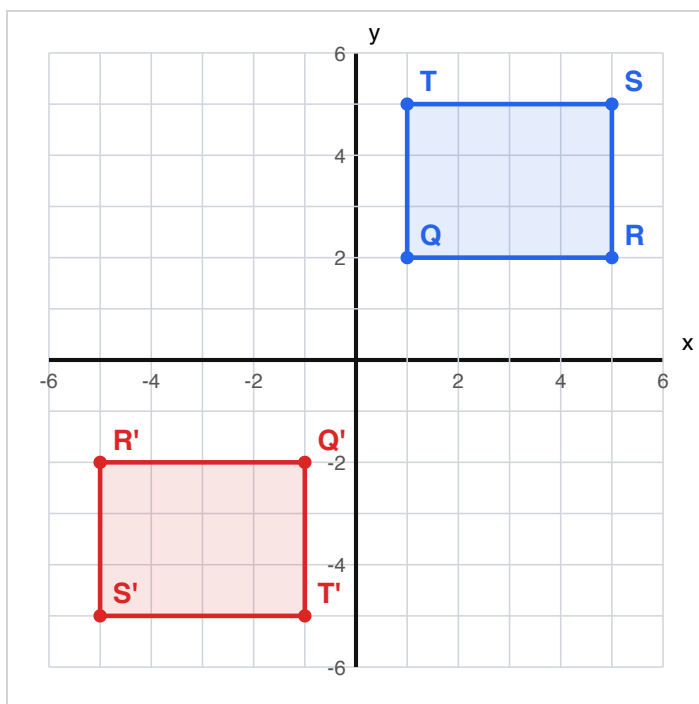
11. Rotate triangle NOP **90° clockwise** about the origin.



N'(1, 5), O'(1, 2), P'(4, 5)

$(x, y) \rightarrow (y, -x)$.

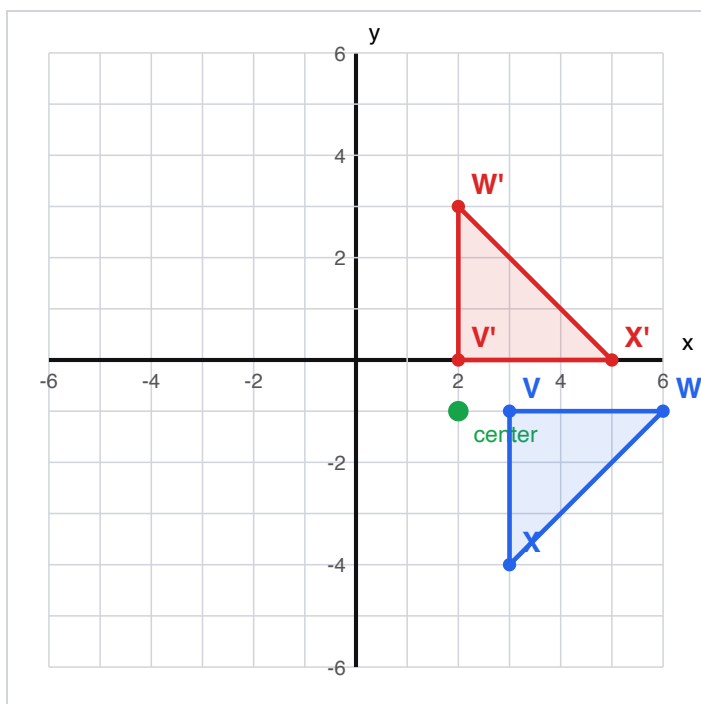
12. Rotate square QRST **180°** about the origin.



Q'(-1, -2), R'(-5, -2), S'(-5, -5), T'(-1, -5)

$(x, y) \rightarrow (-x, -y)$.

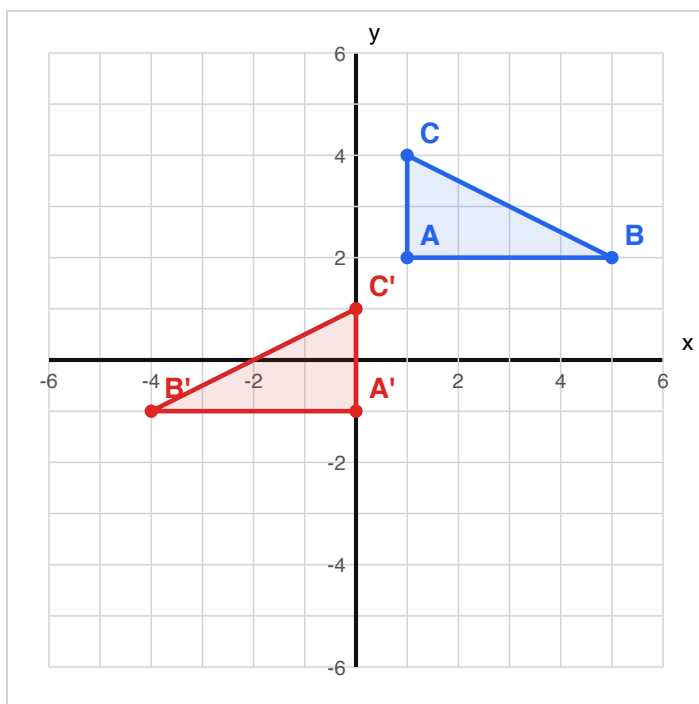
13. Rotate triangle VWX **90° counterclockwise** about the point (2, -1), **not** the origin.



V'(2, 0), W'(2, 3), X'(5, 0)

Subtract the center: (1,0), (4,0), (1,-3). Rotate with $(x,y) \rightarrow (-y,x)$: (0,1), (0,4), (3,1). Add the center back: (2,0), (2,3), (5,0).

14. Triangle ABC: **first** reflect over the y-axis, **then** translate by $\langle 1, -3 \rangle$.

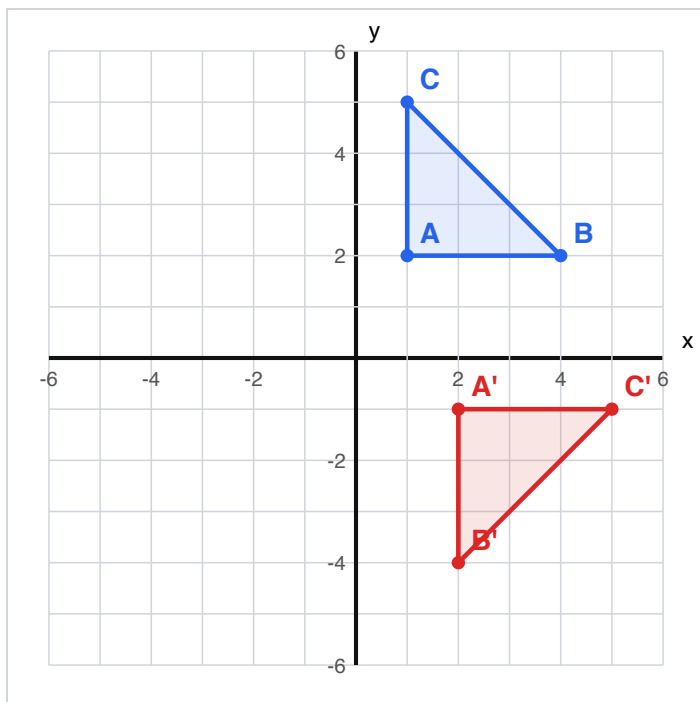


A'(0, -1), B'(-4, -1), C'(0, 1)

After the reflection: (-1,2), (-5,2), (-1,4). After the translation: (0,-1), (-4,-1), (0,1).

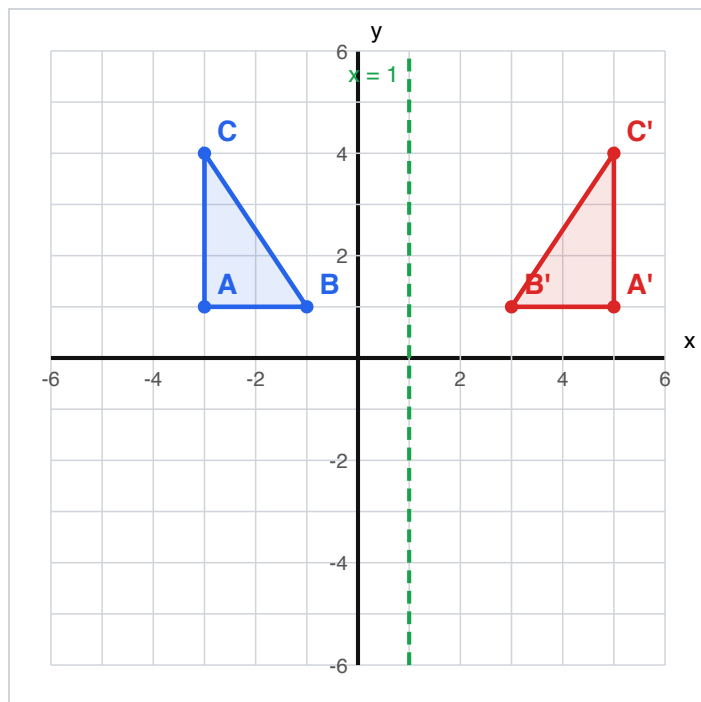
Part 3: Describe the Transformation (2 points each)

15. Describe the transformation that maps triangle ABC to A'B'C'. Be specific (direction, angle, center).



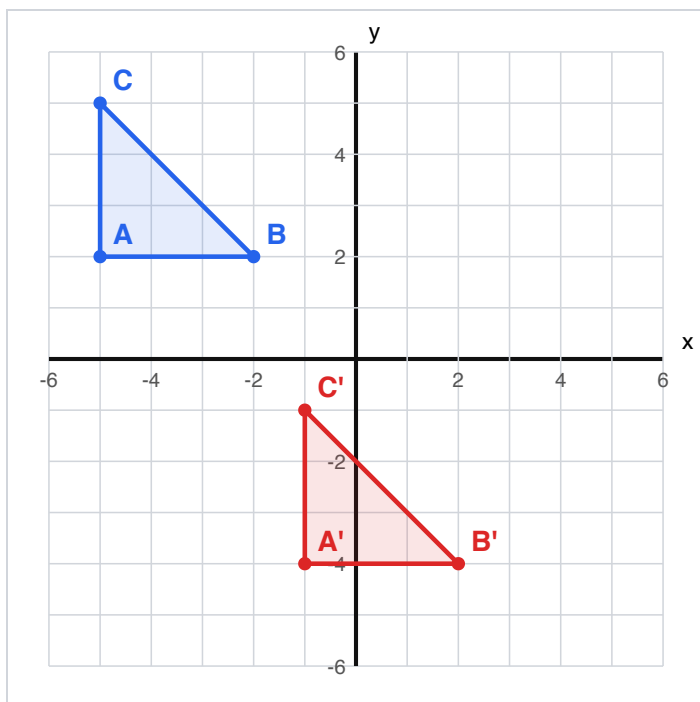
Rotation of 90° clockwise about the origin (same as 270° counterclockwise).

16. Describe the transformation that maps triangle ABC to A'B'C'. Give the equation of the line.



Reflection over the vertical line $x = 1$ (the line halfway between $x = -3$ and $x = 5$).

17. Describe the transformation that maps triangle ABC to A'B'C'. Give the vector.



Translation by $\langle 4, -6 \rangle$.

Part 4: Congruence and Reasoning (2 points each)

18. Triangle ABC has A(0, 0), B(3, 0), C(0, 4). Triangle DEF has D(5, 1), E(8, 1), F(5, 5). Describe a translation that maps ABC onto DEF, and use it to explain why the triangles are congruent.

The translation $\langle 5, 1 \rangle$ sends $A \rightarrow (5, 1) = D$, $B \rightarrow (8, 1) = E$, $C \rightarrow (5, 5) = F$. A translation is a rigid transformation, so it preserves lengths and angles. The triangles are congruent.

19. Triangle PQR has P(1, 1), Q(4, 1), R(1, 3). Triangle P'Q'R' has P'(2, 2), Q'(8, 2), R'(2, 6). Could a rigid transformation map PQR onto P'Q'R'? Use side lengths to explain.

No. $PQ = 3$ but $P'Q' = 6$. Rigid transformations preserve distance, so they can't change a side from 3 to 6. This is a dilation by a factor of 2, so the triangles are similar but not congruent.

20. The point (4, -3) is rotated 90° counterclockwise about the origin and then reflected over the x-axis. Find the final point. Would doing the two steps in the opposite order give the same point? Show work.

Rotate: $(4, -3) \rightarrow (3, 4)$. Reflect over the x-axis: $(3, -4)$.

Opposite order: reflect first $(4, 3)$, then rotate $\rightarrow (-3, 4)$. That is a different point, so order matters.

21. Segment AB has endpoints A(-2, 1) and B(2, 4). It is reflected over the y-axis. Find the length of A'B' **without** computing the new coordinates, and explain how you know.

$AB = \sqrt{4^2 + 3^2} = 5$. A reflection is a rigid transformation, so it preserves length. $A'B' = 5$.

22. A(2, 6) maps to A'(2, 0) and B(-3, 5) maps to B'(-3, 1) under a reflection. Find the line of reflection and check it with both points.

The x-values don't change, so the line is horizontal. Midpoint of 6 and 0 is 3, and midpoint of 5 and 1 is 3. Both agree: the line is $y = 3$.

Part 5: Constructions (3 points each: 2 for the construction, 1 for the explanation)

23. Construct the **perpendicular bisector** of XY.

Open the compass wider than half of XY. Draw arcs from X and from Y that cross above and below. Draw a line through the two crossings. Why: each crossing is the same distance from X and Y.

24. Construct a **copy of the angle** at V, with its vertex at W and one side along the new ray.

Draw an arc centered at V crossing both sides, and the same-radius arc centered at W. Measure the chord between the two crossing points on the first arc, and mark that chord on the new arc. Draw the ray from W through that mark. Why: the triangles formed have three equal sides (SSS), so the angles match.

25. Construct the **angle bisector** of the angle at B.

Arc from B to cross both sides. From each crossing draw an arc of equal radius so they intersect inside the angle. Draw the ray from B through the intersection. Why: the two triangles are congruent (SSS), so the angles on both sides of the ray are equal.

26. Construct the line **perpendicular** to the given line through point P.

Draw an arc centered at P that crosses the line in two places. From each crossing draw equal arcs on the other side of the line so they cross. Draw a line from P through that crossing. Why: P and the crossing are both equidistant from the two points on the line, so they lie on its perpendicular bisector.

27. Starting at A, construct a **regular hexagon** inscribed in the circle with center O.

Set the compass to the radius OA. Starting at A, step around the circle marking arcs six times. Connect consecutive points. Why: a radius-length chord forms an equilateral triangle with the center, so each central angle is 60° , and $6 \times 60^\circ = 360^\circ$.